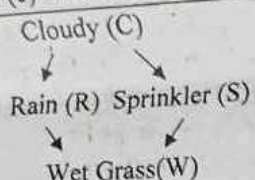


National Institute of Technology Kurukshetra
End Semester Examination, Even Semester 2025-26

Subject: Decision Making Using AI **Subject Code:** AIPC 200 **Branch:** B.Tech. AIML
Semester: IV **Maximum Marks:** 50 **Date:** 03/08/2026 **Time:** 3 Hours
Roll No: _____

Instructions: There are 10 questions in total, and all are compulsory. Please read each question carefully before answering.

1	(a) Explain the following approaches to defining AI: (i) Thinking Humanly, (ii) Acting Humanly (iii), Thinking Rationally, and (iv) Acting Rationally (b) An autonomous vehicle must decide whether to stop, slow down, or change lanes when it detects an obstacle. Explain how rational decision making can be applied in this scenario.	2 3																								
2	Two friends are lost in an $N \times N$ maze and would like to meet; they don't care where. In each time step, both simultaneously move in one of the following directions: {NORTH, SOUTH, EAST, WEST, STOP}. They do not alternate turns. You must devise a plan which positions them together, somewhere, in as few time steps as possible. Passing each other does not count as meeting; they must occupy the same square at the same time. Answer following questions: i) Formally state this problem as a single-agent state-space search Define (a) State (b) Goal test (c) Successor function ii) Give a non-trivial admissible heuristic for this problem. iii) Which search algorithm (BFS or DFS) is guaranteed to find the minimum-time meeting plan in this problem? iv) What is the maximum branching factor of the search tree?	5																								
3	What is Uniform Cost Search? Why does Uniform Cost Search always produce an optimal solution even when edge costs are unequal? Explain with a suitable example. Discuss its time complexity and space complexity for a search tree with branching factor b , solution depth d , and maximum depth m .	5																								
4	Consider a game tree of depth 3 where the root node A is a MAX node, its children B, C, and D are MIN nodes, and their children are MAX nodes with terminal utilities given as follows: under B, nodes E, F, and G have leaf values (3, 5, 2), (9, 1, 4), and (6, 0, 7); under C, nodes H, I, and J have leaf values (4, 2, 8), (5, 6, 1), and (7, 3, 9); under D, nodes K, L, and M have leaf values (2, 4, 6), (8, 5, 7), and (1, 0, 3); i) apply the minimax algorithm to compute the value at the root and determine the optimal move for the MAX player, ii) Apply alpha-beta pruning assuming left-to-right node exploration, identify the pruned nodes and compute the final value,	5																								
5	Consider the following heuristic values for a search problem. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Node</th> <th>True Cost to Goal $h^*(n)$</th> <th>$h_1(n)$</th> <th>$h_2(n)$</th> </tr> </thead> <tbody> <tr> <td>A</td> <td>10</td> <td>8</td> <td>10</td> </tr> <tr> <td>B</td> <td>7</td> <td>6</td> <td>8</td> </tr> <tr> <td>C</td> <td>5</td> <td>5</td> <td>6</td> </tr> <tr> <td>D</td> <td>3</td> <td>2</td> <td>4</td> </tr> <tr> <td>Goal</td> <td>0</td> <td>0</td> <td>0</td> </tr> </tbody> </table> (a) Determine whether h_1 and h_2 are admissible heuristics. (b) Which heuristic is more informed? Justify your answer. (c) Which heuristic is expected to expand fewer nodes when used with the A* algorithm? Explain your reasoning.	Node	True Cost to Goal $h^*(n)$	$h_1(n)$	$h_2(n)$	A	10	8	10	B	7	6	8	C	5	5	6	D	3	2	4	Goal	0	0	0	5
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6	Consider the following 8-Queens configuration. Column : 1 2 3 4 5 6 7 8 Row : 1 4 6 8 2 7 5 3 (a) Calculate the heuristic value (number of attacking queen pairs). (b) Evaluate all possible moves for the queen in Column 2 and compute the heuristic value for each resulting state. (c) Which move will be selected by Hill Climbing? Explain.	5
7	 <pre> graph TD C[Cloudy (C)] --> R[Rain (R)] C[Cloudy (C)] --> S[Sprinkler (S)] R[Rain (R)] --> W[Wet Grass (W)] S[Sprinkler (S)] --> W[Wet Grass (W)] </pre> Consider the above Bayesian Network (a) Explain the concept of d-separation. (b) Determine whether the following pairs of variables are independent and why: (i) Cloudy and WetGrass (ii) Rain and Sprinkler (iii) Rain and Sprinkler given Cloudy (iv) Rain and Sprinkler given WetGrass	5
8	For Bayesian Network in the above question, the following probabilities are given: $P(C) = 0.5$ $P(R C) = 0.8, P(R \neg C) = 0.2$ $P(S C) = 0.1, P(S \neg C) = 0.5$ $P(W RS) = 0.99$ $P(W R\neg S) = 0.90$ $P(W \neg RS) = 0.80$ $P(W \neg R\neg S) = 0.01$ Compute $P(R, S)$, and $P(W)$ showing all intermediate calculations.	5
9	Compare Hill Climbing and Simulated Annealing based on: • Search strategy • Acceptance of worse solutions • Local optimum • Exploration capability • Convergence	5
10	Explain the complete working of a genetic algorithm with the 4-Queens problem. Your explanation should include: • Population initialization • Fitness evaluation • Selection • Crossover • Mutation • Termination criterion Show two iterations of genetic algorithm on 4-Queen problem.	5